Probability, R and Computational Finance

Martin Summer and Branko Urosevic

2021-07-15

Introduction

> "What I hear, I forget; What I see, I remember; What I do, I understand"
>
> — Confucius, 551-479 BC

Our lectures are going to teach you basic concepts of probability and their applications in computational finance. Our approach is based on intuition and on understanding concepts by building them with your own hands.

But how can we build abstract concepts such as probability, random phenomena and chance by our own hands? How can we "touch" such things as ideas? While probability is a mathematical theory, it gains practical value and an intuitive meaning in connection with real or conceptual experiments such as, tossing a coin, rolling two dice, playing roulette, observing the change in the index value of stocks, bonds or in exchange rates, selecting a random sample of financial transactions in a payment system and observing the number of transactions with a value below a certain amount.

Many of these experiments we can nowadays do not only physically – by actually rolling a pair of dice, for example – but we can use the computer to build a virtual dice and roll it on the computer. We can build and simulate a huge variety of random phenomena. We can - for instance - implement models of random fluctuations of asset prices. We can model financial risks and contemplate possible future scenarios through simulation.

We can approach our understanding of abstract concepts by building them with our own hands on the computer. This is the approach to teaching you probability in this course.

There are many ways to build virtual objects and to run simulations to manipulate them. For this we will need a programming language. The language we choose for this course is R and the integrated development environment RStudio. This is one of the main languages used in data analysis, statistics and data science and is widely used in industry and academia. It will be our tool to do probability in this course.

Downloading and installing R

So lets start by downloading and installing R first. R is an open source project maintained by an international team of developers. The software is made available through a website called the comprehensive R archive network (http://cran.r-project.org).

At the top of this website, in a box named "Download and install R" you will find three links for downloading R. Choose the link that describes your operating system, Windows, Mac or Linux. These links will lead you to the necessary information you need to install a current version of R. The easiest install option is to install R from precompiled binaries. There is also the option to built R from source on all operationg systems if you have the tools and the expertise to do so. R also comes both in a 32-bit and a 64-bit version. It does not make a substantial differecne which version you use. 64-bit versions can handle larger files and datasets with fewer memory management problems.

Downloading and installing RStudio

RStudio is an application that helps you write and develop R code. It makes using R much easier fro you than using it in isolation. The interface of RStudio looks the same across all operating systems.

You can download RStudio for free from https://www.rstudio.com/products/rstudio/. Select the box RStudio Desktop and follow the download instructions. RStudio Desktop is free. Note that you need to have a version of R installed to use RStudio.

If you have successfully installed R and RStudio, we are ready to start.

Empirical background and some terminology

If you chose to study Finance it should be natural for you to study probability. After all, Finance studies the allocation and pricing of money across time. You can safe money to use it later for purchasing goods and services you can borrow money to invest today and repay in the future from the revenues of your project. But the future is uncertain and we do not know today how it will look like. In Finance we unavoidably have to deal with risk and uncertainty. The most fundamental concept we have available to deal with risk and uncertainty and to think about and analyze it is probability. Therefore every serious student of Finance has to study probability early on.

Probability is a mathematical theory. This theory gets practical value and intuitive meaning in connection with real or conceptual experiments such as rolling a dice, tossing a coin or selecting a random sample of stocks and observing their most extreme price drop over a given period of time or observing the frequency of car accidents etc. All of these descriptions are, however, rather vague. We have to be more precise what we mean by an experiment or observation.

When we use the term (random) *experiment* we mean a process leading to an uncertain outcome. To make this notion precise and useful we need to agree what we mean by uncertain outcomes.

Take the simple example of tossing a coin. In a practical situations the outcome of a coin toss can be that the coin lands on head or tail. But it could in principle also fall on an edge, stand and roll away. Still when we think about the experiment of tossing a coin in the theory of probability we agree *on the outset* that heads and tail are the only possible outcomes of this experiment. By such an agreement we have pinned down the possible outcomes.

The collection of all possible outcomes of an experiment is calles the *sample space*. While the sample space of the experiment of tossing a coin is an idealization it is exactly this idealization which simplifies the theory without affecting its applicability.

Idealizations of this kind are standard in probability and are widely used also

in Finance. It is - for example - impossible to measure how high the price of a stock might become. Is there a maximal price beyond which the price can not possibly be higher? Many of us would hesitate to claim that the price might rise without bound. Yet many models in applied Finance are based on such an assumption. The models allow arbitrary price hikes but with arbitrary small probability as the price gets higher and higher. Practically it does not make sense to believe that a security price can become arbitratily high. The use of arbitrarily small probabilities in a financial model might seem absurd but it does no practical harm and makes the model simple and convenient to use. Moreover, if we seriously introduced an upper bound on a security price at x it would be also awkward to assume that it is impossible that it could be just a cent higher, an assumption equally unappealing as assuming it can get in principle arbitrarily high.

When we talk of a *basic outcome* we mean the possible outcome of a random experiment. An *event* is a subset of basic outcomes. Any event which consists of a single outcome in the sample space is called a *simple event*.

Probability is a measure of how likely an event of an experiment is. Thus when we talk about probability in a precise and meaningful sense we can only do so in relation to a given sample space or to a certain conceptual experiment.

Thus before we can begin to seriously talk about probability we need to develop a good understanding of experiments, sample spaces, basic outcomes and events. In the spirit of this course we will develop this understanding by building these concepts using R. This will at the same time get us started with using the R language as a tool which will allow us to "touch" probability concepts. As we proceed through our course we will develop a broader and deeper understanding of this language together with the probability concepts.

Building experiments, sample spaces and events using R

Lets start with a classic and old example of a probability model: Rolling a six sided die. We want to roll this die not physically but on our computer using R and RStudio as our tool. How can we doe this?

The R User Interface

Before we can ask our tool to do anything for us, we need to know how to talk to it. In our case RStudio allows us to talk to our computer. It works like any other application. When you launch RStudio on your computer, you will see a screen looking like this:

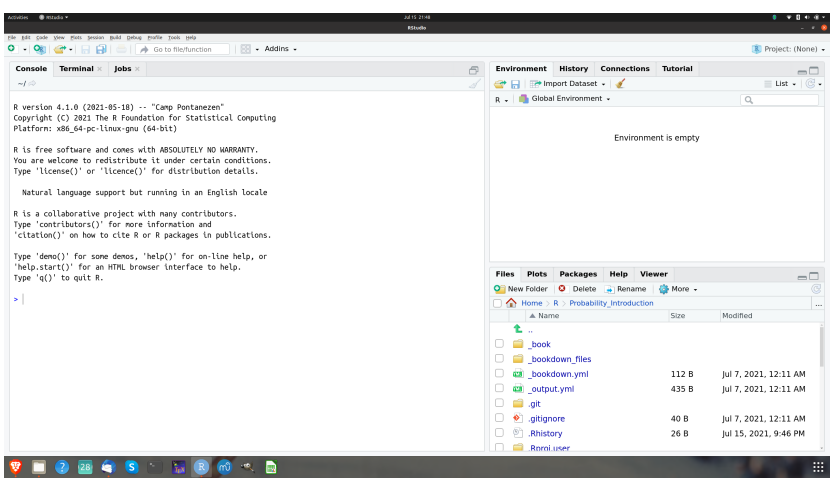

Figure 1: The RStudio startup screen

In this picture you see a creenshot of my RStudio screen. Interacting with the app is easy. You type commands via your keyboard at the prompt, which uses the `>` symbol which you can see on the left in the screenshot. You send the command to the computer by pressing enter. After you have pressed enter, RStudio sends the command to R and displays the result of your command with a new prompt to enter new commands, like this:

```
> 1 + 1
[1] 2
>
```

The `[1]` means that the line begins with the first value of your result. For example, if you enter the command `20:60` which means in the R language list all the integers from 20 to 60, and press return, you will get an output like this:

```
[1] 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 4
[34] 53 54 55 56 57 58 59 60
```

meaning that 20 is the first value displayed in your result. Then there is a line break because not all values can be displayed on the same line and R tells you that `53` is the 34-th value of the result.

The colon operator `:` is a very useful function in R which we will often need. It allows us to create sequences of every integer between two integers.

R needs a complete command to execute it, when the return key is pressed. If a command is incomplete, like for instance `5*`, where we enter 5, the multiplication operator but no second factor to multiply 5 with R will show:

```
> 5*
+
```

The `+` sign means that R needs more input to complete the expression. So if we type the missing factor after the `+` prompt, say 4 and then press the return key again, we will get the result

```
> 5*
+ 4
[1] 20
```

If you type a command that R does not understand, you will be returned an error message. Don't worry if you see an error message. It just is a way the computer tells you that he does not understand what you want him to do.

For instance, if you type `5%3` you will get an error message like this

```
> 5%3
Error: unexpected input in "5%3"
>
```

Sometimes it is onbvious that a mistake occurred, in this case that R just does not know what to do with the symbol `%`. It has no meaning in this context.

Sometimes it is not so obvious what the error message actually means and what you might do about it. A usuful strategy in this case is to type the error message into a seach engine and see what you can find. The chance is very high that others encountered the same problem before you and got helpful advice how to fix it from other users on the internet. One site, we find particularly helpful for all kinds of questions related to R and R programming is https://stackoverflow.com/. Try it at the next opportunity.

Now with this basic knowledge, we can already make the first step to create a die on the computer using R. If you think of a physical die, the essential thing that matters are the points on its six sides. If you throw the dice it will usually land on one of these sides and the side which is on the upper side of the die shows the number of points. The colon operator `:` gives us a way to create a group of numbers from 1 to 6. R gives us the result as a one dimensional set of numbers.

```
> 1:6
[1] 1 2 3 4 5 6
>
```

Lets use these first steps in R to recap the probability concepts we have learned using this example of the six sided die: A *basic outcome* of rolling a six-sided die is for example `6` if the upper side after rolling the die happens to be the side with 6 points. The *sample space* of the experiment of rolling a six-sided die is the set $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. In probability theory we often use the symbol $\mathcal{S}$ or S for sample space. In many probability texts the sample space is also often denoted by the symbol Ω the greek letter for (big) Omega. A *random experiment* in this example is the rolling of the die. The outcome is uncertain but once the die is rolled the outcome can be determined precisely. Finally we can also illustrate in the example the concept of an *event*: For example, the event, that the die will show less than 4 points is $A = \{1, 2, 3\}$. We use here the symbol A to denote this event and use set notation to describe the set of basic outcomes that make up the event "number of points smaller than 4." The event that the outcome is that the point shows 10 points is the empty set $B = \emptyset$. The symbol $\emptyset$ comes from set theory and means the set containing no elements. This event can contain no elements because we can not get a score of 10 by rolling a six sided die.

Lets go back to R now.

Objects

Literature

Here is a review of existing methods.

Methods

We describe our methods in this chapter.

Applications

Some *significant* applications are demonstrated in this chapter.

Example one

Example two

Final Words

We have finished a nice book.

Appendix

Installing R

Installing RStudio